ECON 508 PSM

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Bellman Equation

- Objective

$$\Pi[p] = E\Sigma\beta^t \underset{t=1}{\overset{\infty}{\pi}}(p_t)$$

it can be writen into

$$V(x_1) = \max_{\{p_t\}_{t=1}^{\infty}} \sum_{t=1}^{\infty} \beta^{t-1}\pi(p_t)$$
$$s.t. p_t \in P(x_t), p_{t+1} = T(p_t, x_t), \forall t = 1, 2, ...$$

- Bellman Principle

$$V(x_1) = \max_{p_1} \{\pi(p_1) + \beta V(x_2)\}$$
$$s.t. p_1 \in P(x_1), p_2 = T(p_1, x_1)$$

where

$$V(x_2) = \max_{\{p_t\}_{t=2}^{\infty}} \sum_{t=2}^{\infty} \beta^{t-2}\pi(p_t),$$
$$s.t. p_t \in P(x_t), p_{t+1} = T(p_t, x_t), \forall t = 1, 2, ...$$

Model (one price)

- Objective

$$E(\pi(p))$$

where

$$x = D(p) = exp(b_0 + b_1 log(p) + b_2 log^2(p))$$
$$\pi^*(p) = xp - C(x)$$

- Steps

i) Sample $b = (b_0,\ b_1,\ b_2)$ from its prior distribution;
ii) Find $p^{(i)} = argmax\pi^*(p)$;
iii) Redo i), ii) for n times, get a$p^{(1)},\ p^{(2)},\ \cdots,\ p^{(n)}$;
iv) $p^* = \dfrac{\sum_{i=1}^{n} p^{(i)}}{n}$

Model (one price) for t periods

- Objective

$$\Pi(p) = E(\sum_{i=1}^{t} \beta^i \pi(p))$$

where

$$x^{(i)} = D(p) = exp(b_0^{(i)} + b_1^{(i)} log(p) + b_2^{(i)} log^2(p))$$
$$\Pi^*(p) = E \sum_{i=1}^{t} \beta^i (x^{(i)} p - C(x^{(i)}))$$

- Steps

i) Sample $b^{(i)} = (b_0^{(i)}, b_1^{(i)}, b_2^{(i)})$, $i = 1, \cdots, t$ from its prior distribution;
ii) Find $p^{(j)} = argmax \prod^*(p)$;
iii) Redo i), ii) for n times, get a$p^{(1)}, p^{(2)}, \cdots, p^{(n)}$;
v) $p^* = \dfrac{\sum_{i=1}^{n} p^{(i)}}{n}$

Model ($\beta = 1$)

- Objective

$$\Pi[p] = E \sum_{t=1}^{\infty} \pi(p_t)$$

- Steps
  i) Choose a initial price p_0;
  ii) Construct an price interval based on previous price:$(p_t + h, p_t - h) \forall t = 0, 1, 2, ...$for some arbitrary positive number$h$;
  iii) Pick up a price in the price interval that maximizes the profit as our next period price$p_{t+1} : p_{t+1} = argmax \pi(p), s.t.\ p \in (p_t + h, p_t - h)$;
  iv) Continue choosing prices for remaining periods like above and constitute the sequence of prices $p_t : t = 0, 1, 2, ...$

Model

- Objective

$$\prod[p] = E \sum_{t=1}^{\infty} \beta^t \pi(p_t)$$

where β is the discount factor

$$\beta = \frac{1}{1+r(month)}$$
$$r(month) = (1 + r(annual))^{1/12} - 1$$
$$\beta = \frac{1}{(1+r(annual))^{1/12}}$$

Algorithm

- Iteration

$$p_{t+1} = argmax\, E \sum_{i=1}^{t+1} \beta^i \pi(p_i)$$
$$p_{t+1} \in (min\{0, p_t - c_t\},\ p_t + c_t)$$

- Steps
  i) Given initial point p_0, get $\pi(p_0)$;
  ii) Choose $p_{t+1} \in (min\{0, p_t - c_t\},\ p_t + c_t)$, calculate $s = \sum_{i=0}^{t+1} \beta^i \pi(p_i)$, and record $D(p_{t+1})$;
  iii) Find $p_{t+1} = argmax\{s\}, andD(p_{t+1})$
- Estimate the demand curve using $p_t, D(p_t)$

$$log(D(p)) = b_0 + b_1 log(p) + b_2 log^2(p)$$

Kiefer-Wolfowitz

- Iteration

$$p_{t+1} = p_t + a_t(\pi(p_t + c_t) - \pi(p_t - c_t))/2c_t)$$

- Searching for a sequence

$$p_{t+3} = p_t + a_t(\pi(p_{t+1}) - \pi(p_{t+2}))/2c_t$$

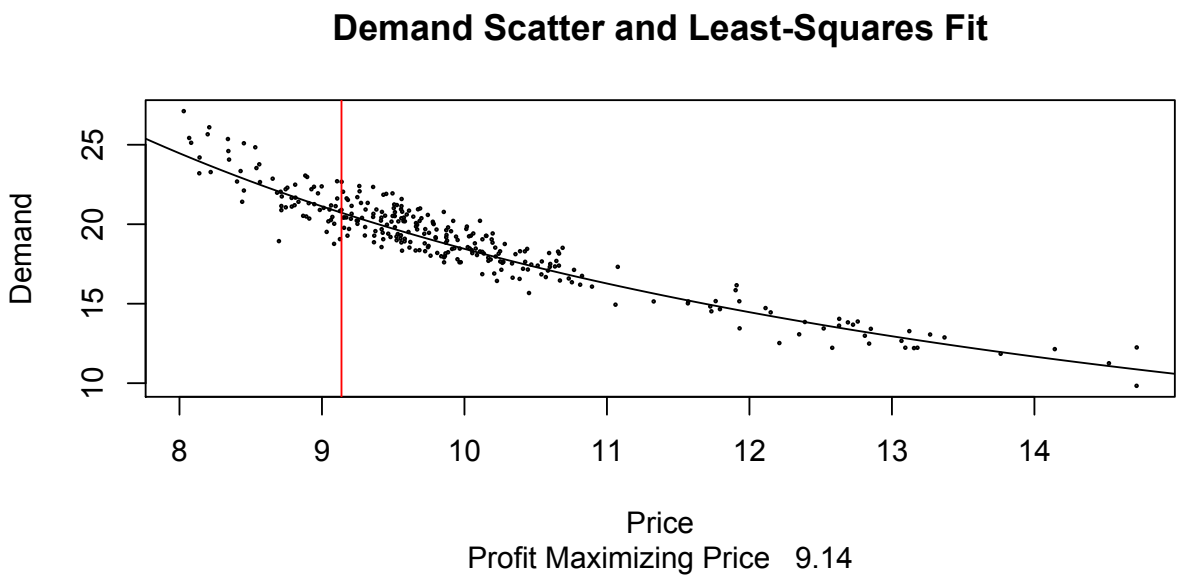
Results

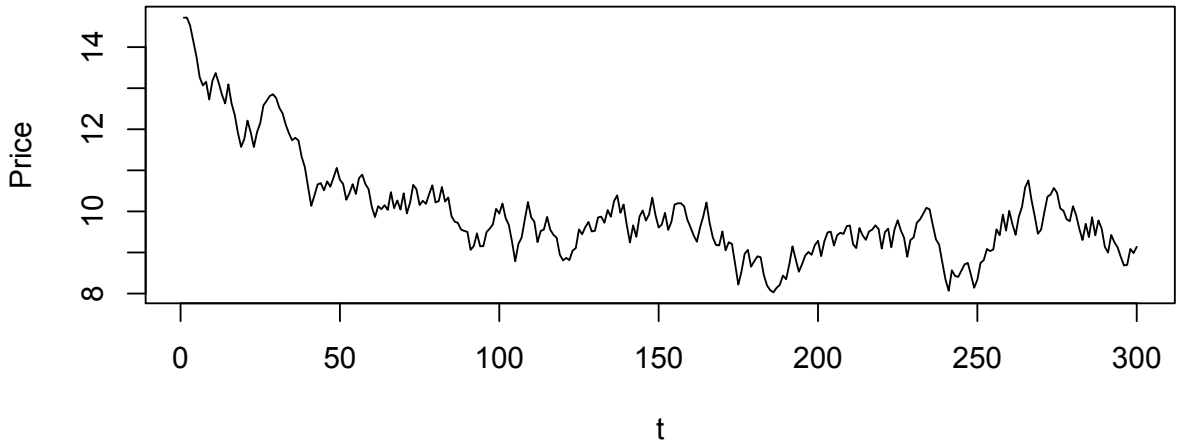
- Demand Curve & Discounted Profit

b	A	p-value	KW	p-value
b_0	4.9979 (0.8263)	4.39e-09	4.93114 (0.50735)	¡2e-16
b_1	-0.5063 (0.7017)	0.471	-0.44113 (0.43487)	0.3116
b_2	-0.1730 (0.1487)	0.246	-0.18738 (0.09274)	0.0447
Profit	50407.4		42246.69	

- Number of Period=300

Results


Demand Scatter and Least-Squares Fit
Demand
10
15
20
25
8
9
10
11
12
13
14
Price
Profit Maximizing Price 9.14


Price
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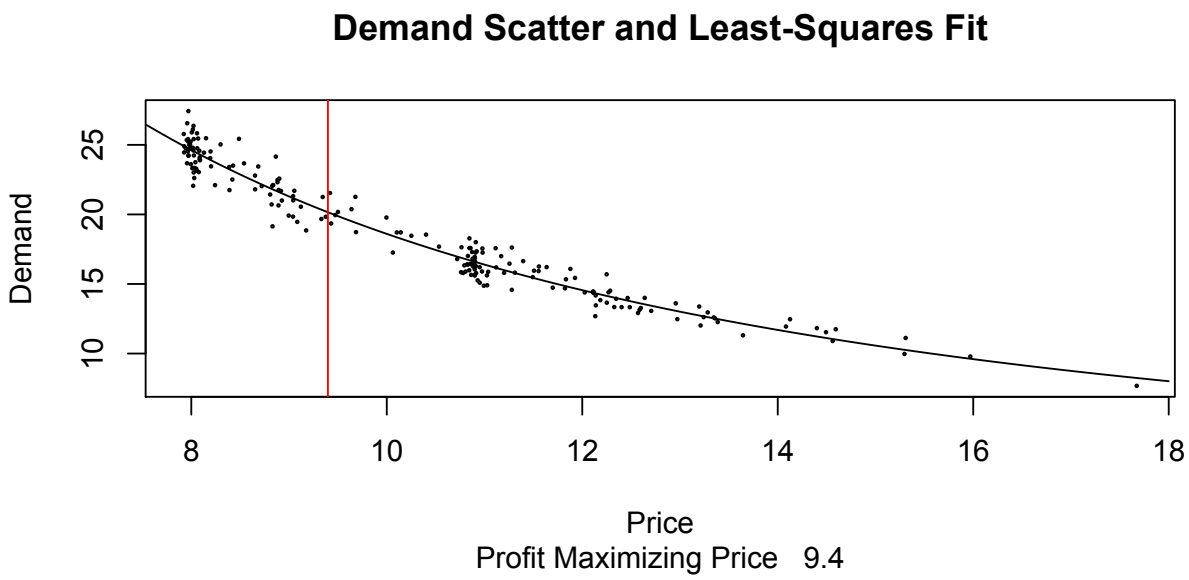
Results


Demand Scatter and Least-Squares Fit
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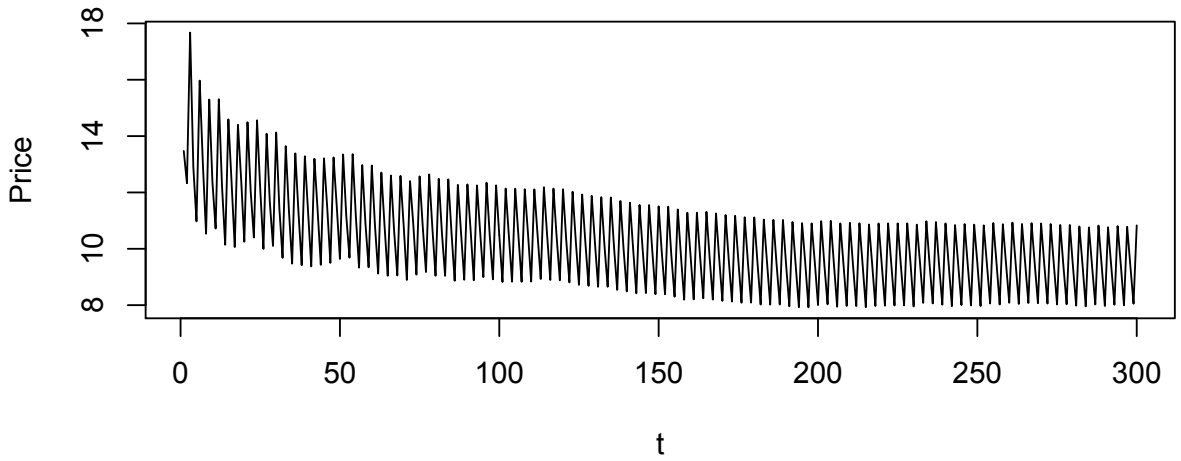
Price
Profit Maximizing Price 9.4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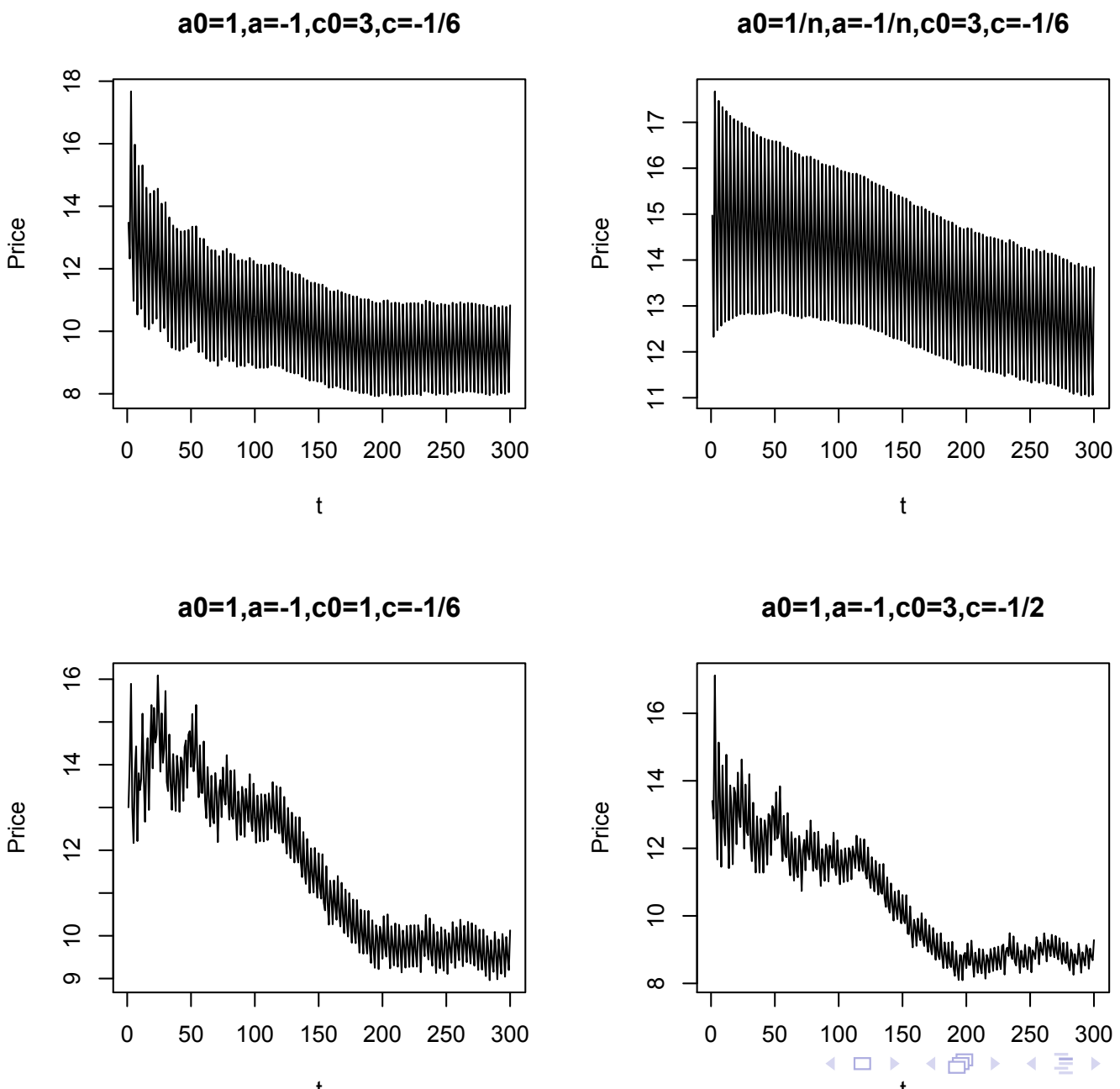
Tuning values for KW



Tuning values for KW

- If c, c_0 hold, the speed of convergence depends more on the value of a, a_0, which determine the length of each step
- Holding a, a_0, c, the speed of convergence does not change a lot, but the fluctuation between two successive periods is smaller as c_0 decreases
- If $c$is changed, the speed of convergence does not change a lot, but the fluctuation between two successive periods is smaller as c decreases

N-dimensional KW

- Iteration

$$\tilde{p}_{t+1} = \tilde{p}_t + \tilde{a}_t(\pi(\tilde{p}_t + \tilde{c}_t) - \pi(\widetilde{p}_t - \tilde{c}_t))/2\tilde{c}_t$$

where $\tilde{p}_t$, $\tilde{a}_t$, $\tilde{c}_t$are N-vectors, the choice of tuning values can be different for each period depending on the speed of convergence.